

6	(a) (i) chemical to electrical	B1	[1]
	(ii) electrical to thermal / heat or heat and light	B1	[1]
	(b) (i) $(P_B =) EI$ or $I^2(R_1 + R_2)$	A1	[1]
	(ii) $(P_R =) I^2 R_1$	A1	[1]
	(c) $R = \rho l / A$ or clear from the following equation	B1	
	ratio = $I^2 R_1 / I^2 R_2 = \frac{\rho l / \pi d^2}{\rho(2l) / \pi(2d)^2}$ or R_1 has $8 \times$ resistance of R_2	C1	
	= 8 or 8:1	A1	[3]
	(d) $P = V^2 / R$ or E^2 / R	C1	
	(V or E the same) hence ratio is $1/8$ or $1:8 = 0.125$ (allow ecf from (c))	A1	[2]